

Research Plan: QIntern 2026

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1. Project Title

Benchmarking QSVT-Based Matrix Inversion with Structure-Aware Block Encodings

2. Project Overview

Quantum singular value transformation (QSVT) [1, 2] is one of the most powerful primitives in quantum algorithms, enabling efficient polynomial transformations of block-encoded matrices. A key application is matrix inversion, which underpins quantum linear system solvers, and is central to applications in partial differential equations, quantum chemistry, and machine learning. However, the practical efficiency of QSVT-based solvers depends critically on the quality of the underlying block encoding: generic dense-matrix encodings are expensive, while structure-aware encodings can dramatically reduce circuit depth and resource overhead [3].

This project studies the end-to-end performance of QSVT-based matrix inversion when equipped with *structure-aware* block encodings. We utilize the explicit block-encoding circuits developed in [4] for discrete Laplacians, which achieve substantially lower gate counts compared to general-purpose approaches, as well as existing phase-angle computation routines for polynomial approximation of x^{-1} . These components are composed into end-to-end inversion pipelines for two QSVT variants:

- **Standard QSVT** [1, 2], which applies polynomial transformations to singular values of block-encoded matrices subject to definite-parity and real-coefficient constraints, and
- **Generalized QSVT (GQSVT)** [5, 6], which lifts parity restrictions via general $SU(2)$ signal-processing rotations.

The expected contributions of this project are: (i) complete end-to-end Qiskit implementations of QSVT and GQSVT matrix inversion pipelines for structured matrices; (ii) systematic circuit-level benchmarks across problem dimensions, condition numbers, and boundary condition types, with resource estimates transpiled to IBM hardware gate sets; and (iii) simulation-based verification of inversion fidelity, yielding practical guidance on which QSVT variant and encoding strategy best suits a given problem class.

The matrix inversion problem appears across a broad range of application domains that motivate this study: solving discretized Poisson and heat equations, perturbative linear systems in quantum chemistry, etc. The central question driving this project is: *for which matrix inversion instances does each QSVT variant yield the best trade-off between circuit depth, gate count, and approximation error and how does the choice of structure-aware block encoding have an impact on these?*

3. Expected Background and Preparation

Interns are expected to have a solid understanding of basic quantum computing, linear algebra, and familiarity with quantum algorithms. Some exposure to signal processing or polynomial approximation theory is a plus. A working knowledge of Python and prior experience with Qiskit or another quantum programming framework is necessary. As preparation, interns are encouraged to review the QSVT tutorial in [2], skim the generalized QSP construction in [5], and familiarize themselves with the block-encoding circuits in [4].

4. Work Plan (1 July – 15 August)

Week	Milestones
Week 1-2	Survey the theoretical landscape: review QSVT and GQSVT frameworks [2, 6], the matrix inversion polynomial approximation problem, and the structure-aware block encoding of [4].
Week 3-4	Assemble the qiskit code for end-to-end standard QSVT and GQSVT matrix inversion pipeline. Verify inversion fidelity on Laplacian instances via statevector simulation and confirm agreement with theoretical polynomial approximation error bounds.
Week 5-6	Conduct systematic benchmarks: vary matrix dimension N , condition number κ , boundary condition type, and grid dimension (1D/2D/3D). Record circuit resources and accuracy. Identify regimes where GQSVT outperforms standard QSVT and vice versa. Finalize the open-source codebase and documentation. Draft the benchmarking paper.

5. Deliverables

- An open-source Qiskit codebase providing end-to-end QSVT and GQSVT matrix inversion pipelines for structured matrices.
- A arXiv paper presenting a systematic comparative analysis of standard and generalized QSVT for matrix inversion across problem classes.

6. Guidance and Extension

I will be actively involved in both the theoretical and implementation aspects of the project, providing regular support via Discord to address any queries. Weekly progress meetings will be held to review milestones, with additional ad hoc meetings upon request. A short introductory presentation on QSVT and block encodings, can be arranged at the start of the internship.

If bandwidth permits, several directions for novel extensions can be explored. One promising direction is adaptive QSVT polynomial construction, where coarse spectral estimates obtained during execution are used to dynamically tune the inversion polynomial degree instead of relying on worst-case condition-number bounds. Such adaptivity could substantially reduce circuit depth for structured matrix families whose effective spectral properties are significantly better than pessimistic theoretical estimates.

Interns are also encouraged to propose their own ideas.

References

- [1] András Gilyén, Yuan Su, Guang Hao Low, and Nathan Wiebe. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics. In *Proceedings of the 51st Annual ACM Symposium on Theory of Computing (STOC)*, pages 193–204. ACM, 2019.
- [2] John M. Martyn, Zane M. Rossi, Andrew K. Tan, and Isaac L. Chuang. Grand unification of quantum algorithms. *PRX Quantum*, 2(4):040203, 2021.
- [3] Parker Kuklinski, Benjamin Rempfer, Justin Elenewski, and Kevin Obenland. Efficient block-encodings require structure, 2025.
- [4] Alexandre Boutot and Viraj Dsouza. Explicit block encodings of discrete Laplacians with mixed boundary conditions. 2026.
- [5] Danial Motlagh and Nathan Wiebe. Generalized quantum signal processing. *PRX Quantum*, 5(2):020368, 2024.
- [6] Christoph Sünderhauf. Generalized quantum singular value transformation. 2023.